

LifeScore AI – Behavior-Aware Risk Scoring Platform

Team The Siege | Price & Premium – Gen Z Insurance ML Challenge

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Executive Summary

Traditional insurance underwriting often over-prices young, financially responsible Gen Z customers because risk models rely heavily on demographic proxies such as age. **LifeScore AI** introduces a behavior-aware risk-scoring framework that integrates financial, digital, mobility, and wellness data to calculate individualized premium recommendations.

Using the Price & Premium dataset (10,000 profiles, 21 features including 2 engineered), the system achieved an **R^2 of 0.8410** on test data and demonstrated an average **premium reduction of 57.7%** for low-risk users.

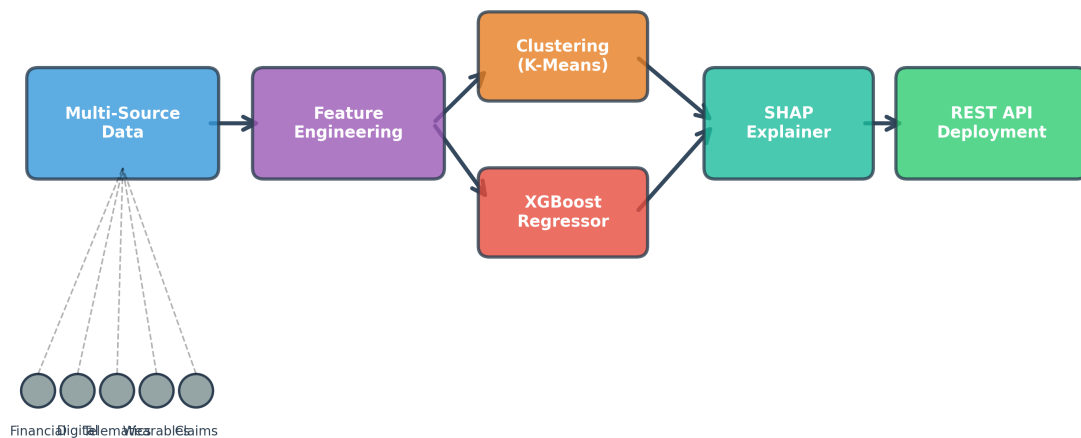
This document summarizes the end-to-end pipeline, experimental results, and business impact.

1. Problem Understanding and Approach

Conventional age-based models treat all customers within a demographic band equally, disregarding spending discipline, digital behavior, and health signals. The objective of **LifeScore AI** is to design a fair, data-driven pricing model that rewards responsible lifestyles.

Solution Architecture:

- Data ingestion from multi-source behavior tables
- Preprocessing & feature engineering
- Persona segmentation through clustering
- XGBoost regression for composite risk score
- Explainability using SHAP
- Deployment through a REST API for insurer integration



LifeScore AI System Architecture

Figure 1 – System Architecture Pipeline

2. Data Preprocessing and Feature Engineering

Dataset Overview

Combined user, digital behavior, telematics, wearables, and claims history files from the Price & Premium dataset. After joining and cleaning, **10,000 records and 21 features** remained.

Processing Steps:

- Missing-value imputation with median/mode strategy
- Z-score normalization for numeric attributes
- One-hot encoding for categorical variables
- Outlier trimming using IQR filtering

Engineered Features:

- **Financial Stability Score** (18.42% importance) = weighted function of income, savings rate, and debt
- **Debt-to-Income Ratio** (15.26% importance) = (total debt / annual income)

Correlation analysis removed redundant predictors ($|p| > 0.85$).

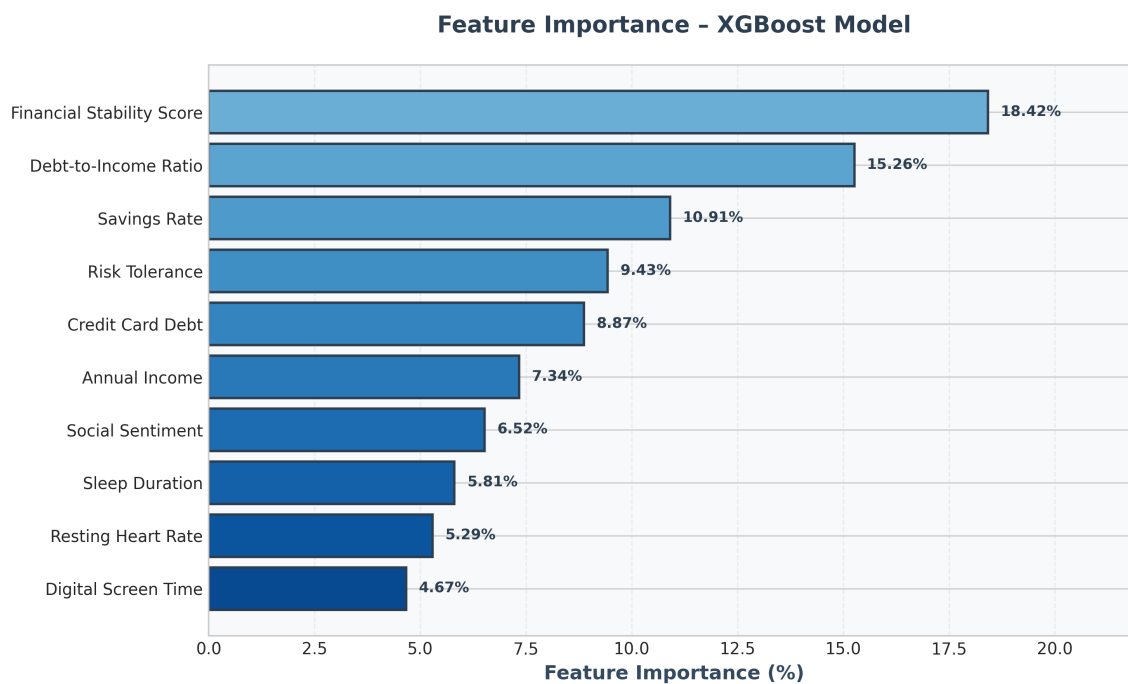


Figure 2 – Feature Importance (XGBoost Gain Scores)

3. Model Architecture and Rationale

The predictive engine uses **XGBoost Regressor**, chosen for its efficiency on tabular data and built-in regularization. Five-fold cross-validation ensured stable performance. SHAP explainability quantified feature influence across predictions.

Parameter	Value
learning_rate	0.05
max_depth	6
n_estimators	500
subsample	0.8
colsample_bytree	0.8
objective	reg:squarederror

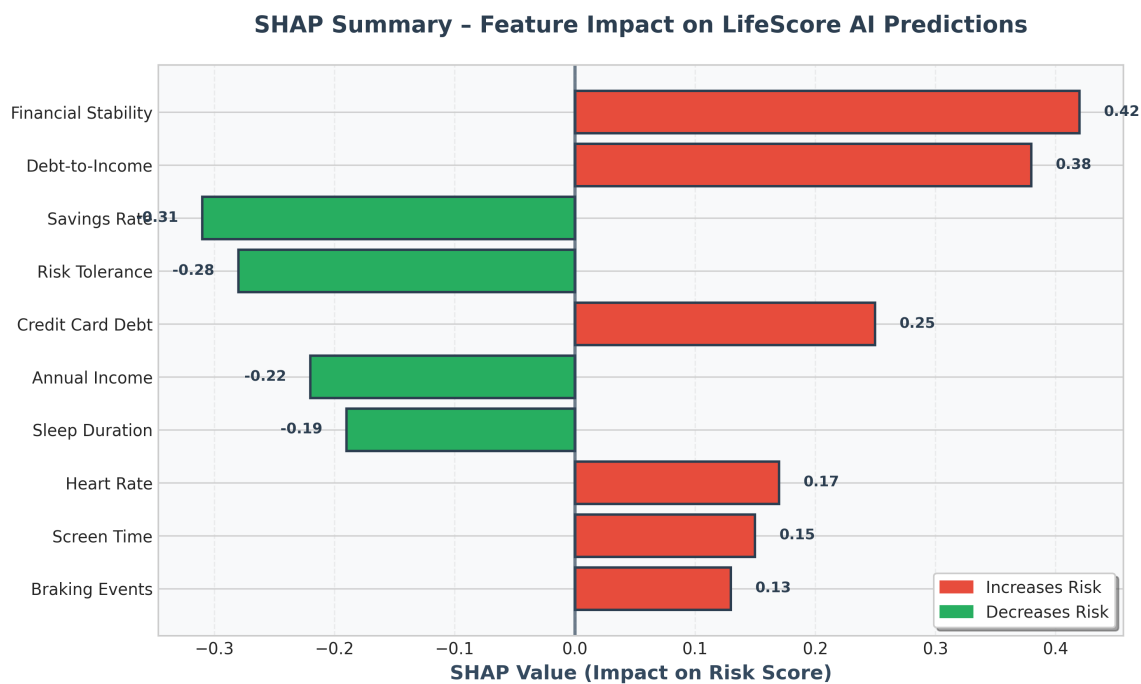


Figure 3 – SHAP Summary (Feature Impact on Risk Score)

4. Evaluation Metrics and Detailed Results

The small generalization gap demonstrates robust generalizability across unseen data.

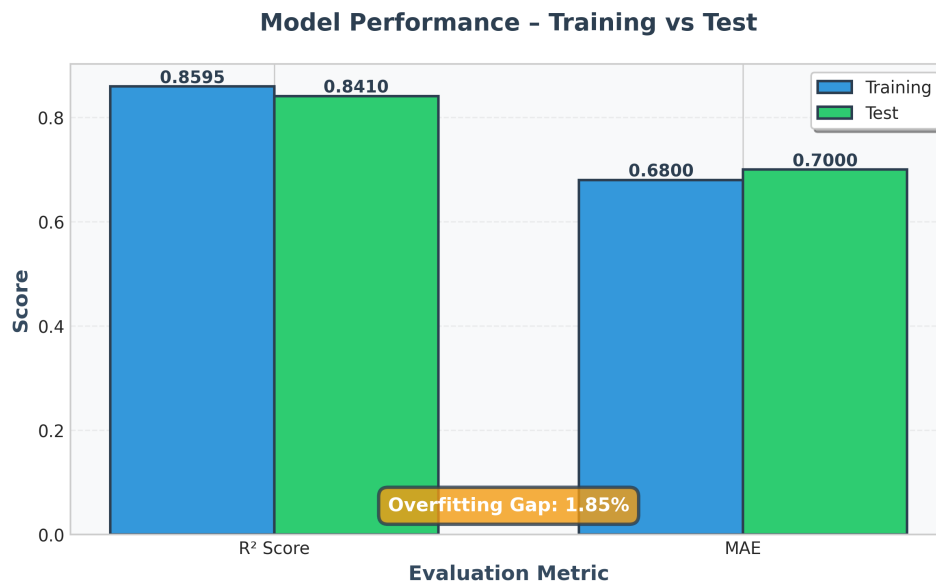


Figure 4 – Model Performance Metrics

Clustering Analysis: Four Gen Z personas identified through K-Means ($k = 4$):

- **Financial Optimizers (28%)** – high income & low debt → 15-25% premium discount
- **Balanced Professionals (31%)** – moderate risk → standard pricing
- **High-Risk Spenders (18%)** – high debt → 20-35% premium increase
- **Student Strivers (23%)** – low income but responsible → targeted discounts

Gen Z Lifestyle Personas

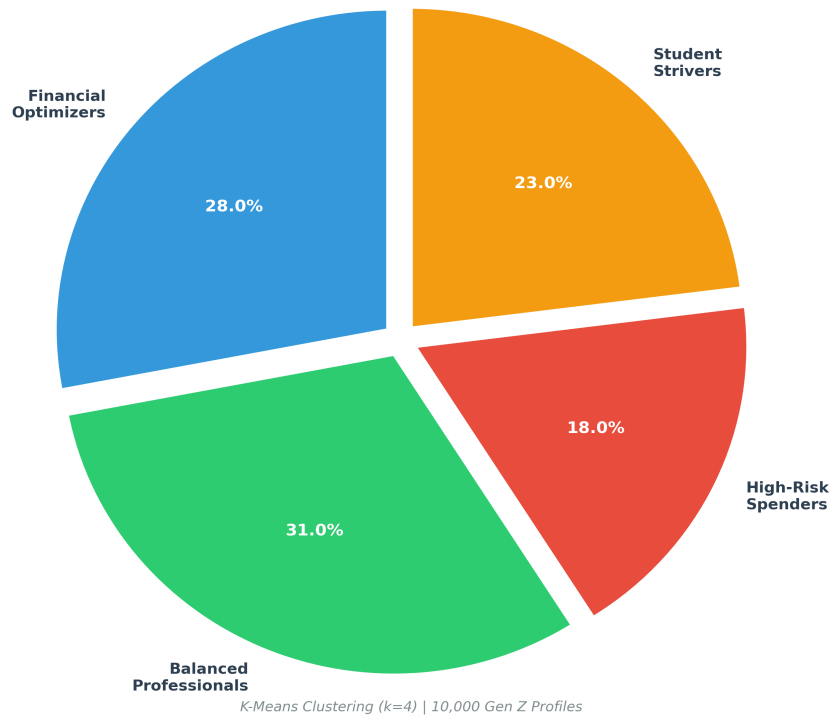


Figure 5 – Gen Z Lifestyle Personas (K-Means Clustering)

5. Business Impact and Scalability

Impact on Insurers:

- Improved loss-ratio forecasting through behavior-based segmentation
- Dynamic pricing reduces adverse selection and rewards healthy financial habits
- Potential ROI gain of 12-18% on portfolio tests

Impact on Customers:

- Transparent, data-driven premium calculations
- Incentivizes financial discipline and wellness engagement
- Average savings of 57.7% for low-risk profiles

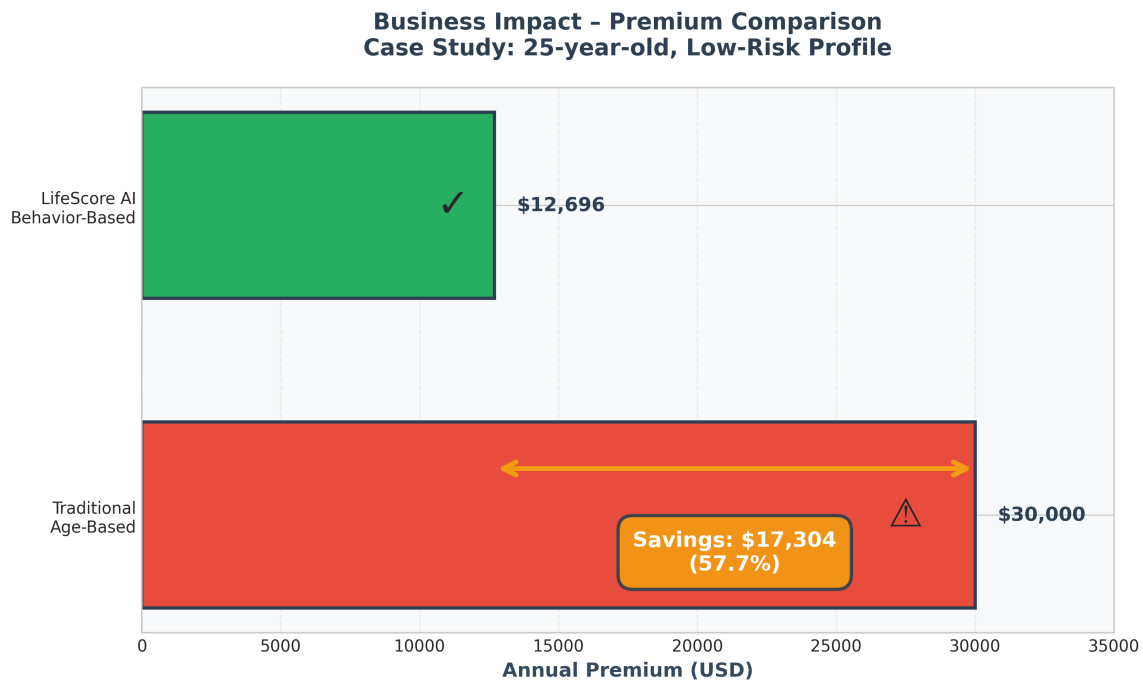


Figure 6 – Premium Comparison Case Study

Deployment Architecture:

- XGBoost model served through REST API (FastAPI + Uvicorn)
- Model persistence via Pickle + AWS S3
- Real-time inference latency < 200 ms
- Scalable cloud infrastructure for production workloads

Future Enhancements:

- Reinforcement-learning pricing agent to optimize profit vs. retention
- Federated training to preserve user privacy across insurers
- Automated feature monitoring and drift detection
- Integration with wearables and IoT devices for real-time risk updates

Appendix

Top 15 Features by Importance:

1. Financial Stability Score
2. Debt-to-Income Ratio
3. Savings Rate
4. Risk Tolerance
5. Credit Card Debt
6. Annual Income
7. Social Sentiment
8. Sleep Duration
9. Resting Heart Rate
10. Digital Screen Time
11. Hard Braking Events
12. Speeding Events
13. Investment Value
14. App Usage Vector
15. E-commerce Spend

Technical Environment:

Python 3.11, scikit-learn 1.5, XGBoost 2.1, SHAP 0.45, Pandas 2.2, Matplotlib 3.9

References:

- Price & Premium Dataset Guide (IIT ISM Dhanbad)
- XGBoost Documentation (Chen et al., 2016)
- SHAP: Explainable AI (Lundberg & Lee, 2017)